

Day 12 - Preprocessing Review

Summary of Work Completed

Today, I reviewed the data analysis and preprocessing work completed so far.

Data Quality

The dataset does not contain any missing values.

Out of 8,512 rows, 8,486 rows have duplicate values. These duplicate values were kept because they are valid repeated measurements caused by the limited resolution of the ADC.

The bit column was found to be identical to the labels column. Therefore, it was removed from the input features to prevent the model from directly getting information about the correct output.

Class Balance

The dataset contains approximately:

- 62.5% Label 0
- 37.5% Label 1

This shows a moderate class imbalance, but it is not severe enough to require resampling at this stage.

During model evaluation, I will check precision, recall, and F1-score along with accuracy to understand the performance of both classes.

EDA Results

The EDA performed on Day 10 showed that the two classes are clearly separated based on the received signal value.

Both the histogram and boxplot showed no overlap between Label 0 and Label 1 at the available turbidity condition of 1–2 NTU.

Data Split and Preprocessing

The dataset was divided using a stratified split:

- Training set: 70% (5,958 rows)
- Validation set: 15% (1,277 rows)

- Test set: 15% (1,277 rows)

The class distribution was maintained across all three datasets.

No additional scaling was required because the feature values are already between 0 and 1.

No label encoding was required because the labels are already in binary format (0 and 1).

Decisions Made

The following preprocessing decisions were reviewed and finalized:

1. Keeping Duplicate Rows

The duplicate rows were kept because they are valid repeated readings from the ADC and are not data entry errors.

2. Removing the bit Column

The bit column was removed because it is identical to the target labels column. Keeping it as an input feature could affect the model results unfairly.

3. No Resampling

No oversampling or undersampling was used because the class imbalance is moderate. The need for resampling will be checked again after evaluating the Week 3 models.

4. No Outlier Removal

No values were removed as outliers because all the signal values appear to be valid physical measurements.

Dataset Version 1

The processed dataset is now finalized as Dataset Version 1.

The files are:

- processed_data/train.csv — 5,958 rows
- processed_data/val.csv — 1,277 rows
- processed_data/test.csv — 1,277 rows

The final preprocessing decisions are:

- No additional feature scaling

- No additional label encoding
- bit column removed from input features
- Stratified train/validation/test split
- random_state=42 used so that the same split can be reproduced

Dataset Version 1 will now be used for the Week 3 model development and experiments.